

Keeper Catch — Emergent Time Is Now in Two Places (the floor's $t = (u+v)/2$ vs the $SO(2)$ clock). The #14+#8 Descent Must Land Exactly (3,1) AND Name ONE Time-Source. Count 8.

Keeper | 2026-07-05 | Consistency catch across Grace's lightcone-floor + Lyra's color→space descent. A budget the descent map must close before either picture banks. Count 8.

Affirm first: Grace's lightcone floor is solid

D_{IV}^5 = tube over the forward lightcone; Jordan algebra = spin factor; rank 2 for any dimension because a lightcone has exactly two null rays. So the rank-2 T^2 floor = the two null circles $u = t+x$, $v = t-x$; $t = (u+v)/2$, $x = (u-v)/2$; left/right movers = a (1+1) 2D CFT (holomorphic $\otimes$ antiholomorphic — the Hardy structure D_{IV}^5 extends). And “can't collapse further” becomes near-theorem: rank 2 is the minimal causal structure (below it, no cone, no direction to commit along). All solid standard geometry (Faraud–Koranyi). The reading of the two circles as time+space with clock-winding = matter/antimatter is the physical layer on top — grounded, not a new theorem. Tiered correctly by Grace.

The catch: two time-sources, and the descent drops the wrong one

Two same-session results assign emergent time to different structures: - Grace: time lives on the floor — $t = (u+v)/2$, and the winding on the floor circle is the commitment tick. - Lyra's descent: drop the 2 floor directions, keep the 3 color directions as space; the floor “stays behind, eternal not-here.”

Naive composition drops time: time on the floor (Grace) + descent drops the floor (Lyra) $\Rightarrow$ emergent matter keeps 3 space, loses its clock. Cannot be intended.

The signature bookkeeping confirms it isn't automatic

$SO(5,2) \rightarrow SO(3,1)$ must drop exactly (2 space, 1 time). Grace's floor is $(1+1) = (1 \text{ space}, 1 \text{ time})$. Dropping the floor removes (1 space, 1 time) — one space short, one time too many. So “drop the floor, keep color = 3 space” does not obviously yield (3,1).

The clean resolution the map should be checked against (not yet shown)

Emergent time = the separate $SO(2)$ isotropy clock, NOT the floor's t . Then: - (3,1) = 3 color ($\rightarrow$ space) + $SO(2)$ clock ($\rightarrow$ time), dropping the rank-2 floor entirely. - Matches Lyra's “floor stays behind” AND lands exactly (3,1). $\square$ - BUT contradicts Grace's “time lives on the floor / winding-on-the-floor = tick.” If emergent time is the $SO(2)$ clock, then the winding-tick, the matter/antimatter sign, and the “eternal shared clock” all live on the $SO(2)$, not on the floor's u,v .

The one question the #14+#8 accounting must answer

Is emergent time the $SO(2)$ isotropy clock, or the floor's $t = (u+v)/2$? They are different structures. The descent (Casey #14 chirality projection + #8 τ -direction) must (a) land exactly (3,1) — 3 space, 1 time — and (b) name a single time-source, not carry two. Whichever it is, the *other* cosmology claims (winding = tick on which circle; matter/antimatter = which winding; eternal shared clock = where) reorganize around it.

Tier

Count 8. Grace's lightcone-floor geometry is solid; the two-time-source tension is a real consistency gap, not a bank-blocker on solid results — it's a gap in the *cosmology interpretation layer* (Level-2) that the descent map must close. Same bolt Lyra already flagged (drop-floor/keep-color must be shown from #14+#8, not assumed);

this catch adds the exact budget (drop (2 space,1 time), keep (3,1)) and the time-source fork that the shown map must resolve.

— Keeper, 2026-07-05. Grace's rank-2 floor = (1+1) lightcone (solid). But time is now claimed both on the floor ($t=(u+v)/2$, winding=tick) and, for a clean (3,1) descent, on the separate $SO(2)$ clock — and the signature budget (drop 2 space+1 time) means “drop the floor” is 1 space short, 1 time over. The #14+#8 descent must land exactly (3,1) and name ONE time-source; the winding-tick and matter/antimatter sign follow it. Count 8.